Day 131 | Unit 09 | Teacher plan

Display, Describe, and Summarize Data

Describe Center, Spread, and Shape

Standards: 6.SP.A.2, 6.SP.A.3

Learning target: I can describe a distribution by center, spread, and shape.

enVision Topic 8: Display, Describe, and Summarize Data | Spectrum companion Chapter 7

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. A distribution shows which values occur and how often. Describe a typical value, how spread out the values are, and whether the shape is roughly symmetric, skewed, clustered, or has unusual values.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 10

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01 / Learn

A distribution shows which values occur and how often. Describe a typical value, how spread out the values are, and whether the shape is roughly symmetric, skewed, clustered, or has unusual values.

Worked example 1

Reading times in minutes are 3, 4, 4, 5, 5, 6. Describe a typical value, range, shape, and any striking value.

1. A distribution shows which values occur and how often.

2. Describe a typical value, how spread out the values are, and whether the shape is roughly symmetric, skewed, clustered, or has unusual values.

3. The values balance around the middle.

4. Range = 6 - 3.

5. State the result in the form requested: Center near 9/2 min; range 3 min; roughly symmetric; no isolated extreme.

Answer: Center near 9/2 min; range 3 min; roughly symmetric; no isolated extreme

Worked example 2

Six students recorded reading times (minutes): 5, 7, 7, 9, 9, 11. Part A: describe center and spread using appropriate measures. Part B: describe shape, count, units, and any striking value; justify the chosen measures.

1. A distribution shows which values occur and how often.

2. Order the observations.

3. Middle positions give median 8 and half medians give IQR 2.

4. Mean is 48/6

5. averaging distances gives MAD 5/3.

6. Tie the chosen summary to the displayed shape and context.

Answer: 6 observations; minutes; median 8, IQR 2; mean 8, MAD 5/3. Roughly balanced; mean/MAD or median/IQR with justification.

Day 131 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Reading times in minutes are 7, 8, 8, 9, 9, 10. Describe a typical value, range, shape, and any striking value.

02 / Constructed response

Reading times in minutes are 5, 6, 6, 7, 7, 8. Describe a typical value, range, shape, and any striking value.

03 / Constructed response

Reading times in minutes are 4, 5, 5, 6, 6, 7. Describe a typical value, range, shape, and any striking value.

04 / Constructed response

Reading times in minutes are 8, 10, 10, 12, 12, 14. Describe a typical value, range, shape, and any striking value.

05 / Constructed response

Reading times in minutes are 3, 8, 8, 13, 13, 18. Describe a typical value, range, shape, and any striking value.

06 / Constructed response

Reading times in minutes are 8, 11, 11, 14, 14, 17. Describe a typical value, range, shape, and any striking value.

07 / Constructed response

Reading times in minutes are 6, 7, 7, 8, 9, 18. Describe a typical value, range, shape, and any striking value.

08 / Statistical reasoning and explanation

Six students recorded reading times (minutes): 6, 8, 8, 10, 10, 12. Part A: describe center and spread using appropriate measures. Part B: describe shape, count, units, and any striking value; justify the chosen measures.

09 / Statistical reasoning and explanation

Six students recorded reading times (minutes): 5, 10, 10, 15, 15, 28. Part A: describe center and spread using appropriate measures. Part B: describe shape, count, units, and any striking value; justify the chosen measures.

10 / Constructed response

Reading times in minutes are 5, 6, 6, 7, 8, 17. Describe a typical value, range, shape, and any striking value.

Day 131 | Unit 09 | Answer key

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Describe Center, Spread, and Shape

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Learning target: I can describe a distribution by center, spread, and shape.

1. Center near 17/2 min; range 3 min; roughly symmetric; no isolated extreme

The values balance around the middle. Range = 10 - 7.

6.SP.A.2, 6.SP.A.3 | Constructed response

2. Center near 13/2 min; range 3 min; roughly symmetric; no isolated extreme

The values balance around the middle. Range = 8 - 5.

6.SP.A.2, 6.SP.A.3 | Constructed response

3. Center near 11/2 min; range 3 min; roughly symmetric; no isolated extreme

The values balance around the middle. Range = 7 - 4.

6.SP.A.2, 6.SP.A.3 | Constructed response

4. Center near 11 min; range 6 min; roughly symmetric; no isolated extreme

The values balance around the middle. Range = 14 - 8.

6.SP.A.2, 6.SP.A.3 | Constructed response

5. Center near 21/2 min; range 15 min; roughly symmetric; no isolated extreme

The values balance around the middle. Range = 18 - 3.

6.SP.A.2, 6.SP.A.3 | Constructed response

6. Center near 25/2 min; range 9 min; roughly symmetric; no isolated extreme

The values balance around the middle. Range = 17 - 8.

6.SP.A.2, 6.SP.A.3 | Constructed response

7. Center near 15/2 min; range 12 min; right-skewed with high value 18 min

The values cluster low with a long high tail. Range = 18 - 6.

6.SP.A.2, 6.SP.A.3 | Constructed response

8. 6 observations; minutes; median 9, IQR 2; mean 9, MAD 5/3. Roughly balanced; mean/MAD or median/IQR with justification.

Order the observations. Middle positions give median 9 and half medians give IQR 2. Mean is 54/6; averaging distances gives MAD 5/3. Tie the chosen summary to the displayed shape and context.

6.SP.A.2, 6.SP.A.3 | Statistical reasoning and explanation

9. 6 observations; minutes; median 25/2, IQR 5; mean 83/6, MAD 11/2. A high value makes median/IQR a useful choice.

Order the observations. Middle positions give median 25/2 and half medians give IQR 5. Mean is 83/6; averaging distances gives MAD 11/2. Tie the chosen summary to the displayed shape and context.

6.SP.A.2, 6.SP.A.3 | Statistical reasoning and explanation

10. Center near 13/2 min; range 12 min; right-skewed with high value 17 min

The values cluster low with a long high tail. Range = 17 - 5.

6.SP.A.2, 6.SP.A.3 | Constructed response